

TELOTROPE

Capability Statement

Resilient AI systems for goal reasoning, simulation, and decision support in high-stakes operational environments.

Telotrope Agents is an R&D firm building resilient AI systems, from autonomous agents to decision-aid tools, for high-stakes operational environments. Our systems reason toward goals, adapt when conditions change, and produce calibrated, decision-relevant output for human operators, using generative models where they earn their place rather than as the reasoning substrate.

Core capabilities

- **Goal reasoning & planning.** Agents that evaluate their own objectives, detect discrepancies against the environment, and re-plan mid-mission without operator intervention.
- **Simulation & adversarial AI.** Rule-bound synthetic environments for combat modeling, wargaming, and strategic stress-testing, including population- and individual-scale socio-cognitive simulation.
- **System diagnosis.** Automated diagnostic reasoning that isolates operational anomalies and produces explainable output, with formal verification and validation of the underlying models.
- **Individual alignment.** Recognizes an individual's goals and intent online, from behavior as it happens, rather than through slow, offline modeling.
- **AI research & advisory.** Basic and applied AI research, evaluation, and technical advisory work scoped to a program's needs, generative or not.

Differentiators

- **Reasoning-first, not generative-only.** Resilient, goal-driven reasoning governs mission-critical decisions; generative models are used selectively, as bounded components for language and perception.
- **Deep technical pedigree.** Leadership has architected and stress-tested this approach across national security and defense R&D programs for two decades.

Company data

NAICS: **541715**, 541511, 541690

CAGE: 22Q64

UEI: FBXRXSVLNCM6

Small business.

Past performance

Matthew Molineaux, CEO: 20+ years in DoD-funded AI R&D, including:

Principal Investigator, DARPA In the Moment (ITM): human-aligned decision-making for high-stakes triage.

Planning Lead, DARPA SCEPTER (Parallax Advanced Research): hierarchical goal planning for machine-generated battle strategy.

Technical Lead, AFRL META (Multi-Sensor Exploitation for Tactical Autonomy) program: support for a broad base of AI initiatives, with some ISR focus.

Selected programs; not a complete list.

Contact

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